LAB 8: Server Configuration - DHCP, DNS and Web Server in Cisco Packet Tracer

OBJECTIVES

- To configure a server to deliver DHCP, DNS, and Web (HTTP) services within a simulated network environment in Cisco Packet Tracer.
- To observe and understand the mechanism by which DHCP distributes IP addresses automatically and how DNS translates domain names into their corresponding IP addresses.
- To test and confirm that all configured services are operational by accessing the hosted web server from connected client machines.

THEORY

1. Server Configuration

A server is a computer that provides services to other devices (clients) on a network. To ensure reliable communication, servers are usually assigned a static IP address. In this lab, one server is configured to run multiple services including DHCP, DNS, and HTTP. These services allow the server to automatically assign IP addresses, resolve domain names, and host web pages. This demonstrates how a single centralized server can efficiently manage and support various network functions.

2. DHCP (Dynamic Host Configuration Protocol)

DHCP is a protocol used to automatically assign network configuration parameters such as IP address, subnet mask, default gateway, and DNS server to devices on a network. Instead of configuring these settings manually on each device, the DHCP server distributes them automatically from a predefined IP address pool. This reduces configuration errors and simplifies network management. In this lab, the server is configured as a DHCP server to automatically provide network settings to connected client PCs.

3. DNS (Domain Name System)

DNS is a system that translates human-readable domain names into numerical IP addresses used by computers. When a user enters a domain name in a browser, the request is sent to the DNS server, which returns the corresponding IP address. The client then uses this IP address to communicate with the destination server. In this lab, a DNS record is created to map the domain name www.hcoe.com to the server's IP address.

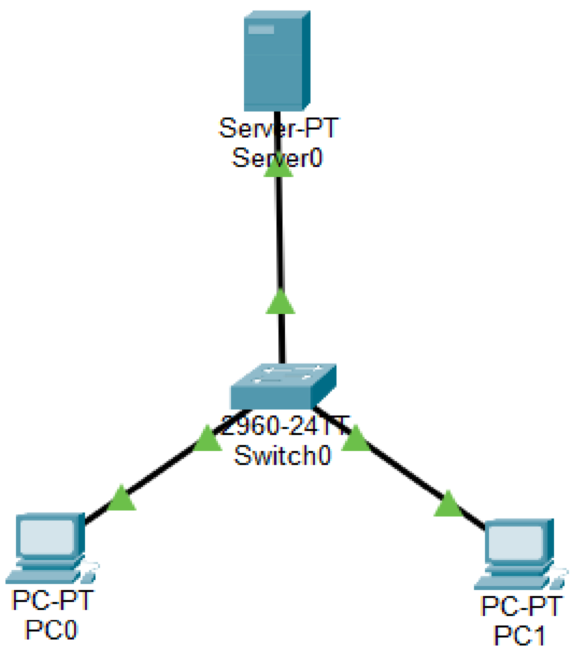
4. Web Server (HTTP Service)

HTTP is a protocol used for transferring web pages between a server and clients through a web browser. When a user enters a website address, the browser sends a request to the web server, which responds by sending the requested webpage. In this lab, the HTTP service is enabled on the server, allowing client PCs to access the hosted webpage through the configured domain name. This demonstrates the basic working of a web server in a local network.

5. Integration of DHCP, DNS, and Web Server

These three services are designed to complement one another. When a PC connects to the network, DHCP first provides it with a valid IP address and the address of the DNS server. When the user then types a web address, DNS resolves that name to the correct IP, and the HTTP service delivers the page. Together, they replicate the core experience of browsing the internet - in a self-contained, controlled lab environment -highlighting the interdependency of foundational network services.

NETWORK TOPOLOGY



CONFIGURATION

IP Addressing Scheme

Overall device addressing:

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

PC configuration after DHCP assignment:

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Server network settings:

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

Procedure

Server Configuration

Step 1: Assign Static IP to Server

1. Click on Server0 and Navigate to Desktop - IP Configuration.
2. Enter the following static values:
   - IP Address: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
   - DNS Server: 192.168.1.2

DHCP Configuration

Step 2: Enable DHCP Service

1. Open Server0, go to Services - DHCP.
2. Switch the DHCP service to ON.

Step 3: Create DHCP Pool

Field	**Value**
Pool Name	LAB POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

After filling in the pool details, click Add to save the configuration.

Verification -DHCP:

On each client PC:

- Open Desktop - IP Configuration and select DHCP mode.
- Confirm that the PC receives a valid IP address from the configured pool.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC0 received 192.168.1.1 and PC1 received 192.168.1.3, both assigned automatically by the DHCP server. Ping tests from PC0 to 192.168.1.3 returned 4 packets sent, 4 received with 0% loss, confirming successful DHCP-based connectivity.

DNS Server Configuration

Step 4: Enable DNS Service

1. Go to Server0 - Services - DNS.
2. Set DNS service to ON.

Step 5: Add DNS Record

- Name: www.hcoe.com
- Type: A record
- Address: 192.168.1.2

Click Add to register the record.

Verification -DNS:

```
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

On each PC, open the Command Prompt and ping www.hcoe.com. The DNS server resolves the domain and returns 192.168.1.2. Replies from both PC0 and PC1 confirmed zero packet loss, with PC1 showing a maximum round-trip of 13ms. This verified that DNS name resolution is working correctly.

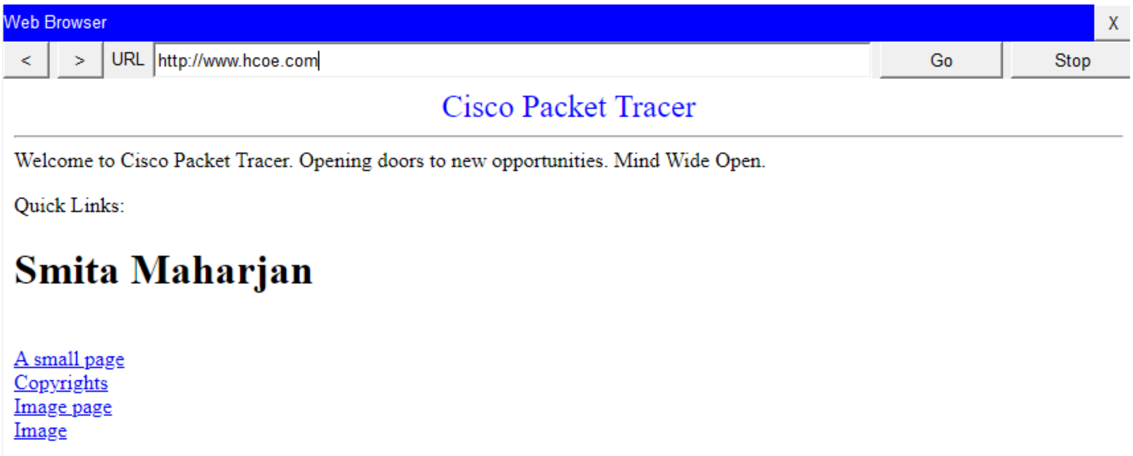
Web Server Configuration

Step 6: Enable HTTP Service

3. On Server0, navigate to Services - HTTP.
4. Turn HTTP to ON.
5. Optionally edit the index.html file to customize the displayed webpage.

Verification -Web Server:

On any client PC, open Desktop - Web Browser and type http://www.hcoe.com. The browser successfully loaded the hosted webpage served by Server0, confirming that the HTTP service is operational and accessible by domain name.



RESULT

All three services -DHCP, DNS, and HTTP -were successfully configured and verified on Server0 in Cisco Packet Tracer. Client PCs obtained network settings automatically through DHCP. Domain name resolution via DNS allowed devices to reach the server by name, and the web server responded to HTTP requests by delivering the hosted page. All configured services operated as expected, meeting the stated lab objectives.

DISCUSSION AND CONCLUSION

This lab provided practical understanding of three key network services: DHCP, DNS, and HTTP. DHCP automatically assigned IP addresses to client devices, DNS translated domain names into IP addresses, and HTTP enabled the server to host and deliver web pages. The lab also showed how these services depend on each other for proper network communication.

All objectives were successfully achieved using Cisco Packet Tracer. Client PCs received IP configurations automatically, resolved the domain name correctly, and accessed the hosted webpage through a browser. Overall, the lab demonstrated the importance of centralized server services in supporting efficient network operations.